

Improved Statistical Inference for Expectile-Based Risk Measures

An Optimal Pooling Approach in the Heavy-Tailed Regime

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Abstract

[TODO:]

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Chapter 1

Introduction

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Chapter 2

Background

2.1 Regular variation and the second-order condition

Throughout the thesis we work with a real-valued random loss X that has continuous distribution function F and survival function $\bar{F} = 1 - F$. Given an iid sample $X_1, \dots, X_n$ from F , the corresponding order statistics are denoted by $X_{1:n} \leq X_{2:n} \leq \dots \leq X_{n:n}$.

Heavy-tailed asymptotics are governed by the language of regular variation; the reference is Haan and Ferreira (2006, Appendix B.1).

Definition 2.1 (Regular variation). A Lebesgue measurable function $f : (0, \infty) \rightarrow \mathbb{R}$ that is eventually positive is *regularly varying at infinity with index* $\alpha \in \mathbb{R}$, written $f \in \text{RV}_\alpha$, if for every $y > 0$,

$$\lim_{t \rightarrow \infty} \frac{f(ty)}{f(t)} = y^\alpha.$$

The functions in RV_0 are called *slowly varying*.

The first-order condition under which all of our asymptotics are taken asserts that F has a Pareto-type tail.

Condition 2.2 (First-order condition). There exists $\gamma > 0$ such that $\bar{F} \in \text{RV}_{-1/\gamma}$, or equivalently the *tail quantile function*

$$U(t) := \inf \left\{ x \in \mathbb{R} : \frac{1}{\bar{F}(x)} \geq t \right\}, \quad t > 1,$$

satisfies $U \in \text{RV}_\gamma$.

Throughout, for a continuous distribution function F , we write

$$Q_X(\tau) := F^{\leftarrow}(\tau) = \inf \{ x \in \mathbb{R} : F(x) \geq \tau \}, \quad 0 < \tau < 1.$$

With the tail quantile $U(t) = F^{\leftarrow}(1 - 1/t)$, $t > 1$, this gives

$$Q_X(\tau) = U((1 - \tau)^{-1}), \quad Q_X(1 - p) = U(1/p).$$

Condition 2.2 is the Fréchet max-domain condition with extreme-value index γ (Haan and Ferreira 2006, Theorem 1.2.1); standard examples include Pareto, Burr, Student- t , Fréchet and F -distribution tails (Haan and Ferreira 2006, §2.6). For normal limits we use the standard second-order refinement.

Condition 2.3 (Second-order condition $\mathcal{C}_2(\gamma, \rho, A)$). There exist $\gamma > 0$, a second-order parameter $\rho \leq 0$, and a measurable auxiliary function A of constant sign with $A(t) \rightarrow 0$ as $t \rightarrow \infty$, such that for every $y > 0$,

$$\lim_{t \rightarrow \infty} \frac{1}{A(t)} \left[\frac{U(ty)}{U(t)} - y^\gamma \right] = y^\gamma \frac{y^\rho - 1}{\rho},$$

where the right-hand side is read as $y^\gamma \log y$ when $\rho = 0$.

The function $|A|$ is itself regularly varying with index ρ (Haan and Ferreira 2006, Theorem 2.3.9). Since A has constant sign eventually, signed ratios $A(ty)/A(t)$ converge to y^ρ ; the rate A is the source of the second-order bias terms below.

Condition 2.4 (Finite first-moment condition). The extreme-value index satisfies $\gamma \in (0, 1)$ and the negative part $X^- = \max(-X, 0)$ satisfies

$$\mathbb{E} X^- < \infty.$$

Under **Condition 2.2**, $\gamma < 1$ gives $\mathbb{E} X^+ < \infty$; together with the displayed lower-tail condition this is the finite-mean regime for the quantile-based expectile approximation.

Condition 2.5 (LAWS lower-tail moment condition). For some $\delta > 0$, the negative part $X^- = \max(-X, 0)$ satisfies

$$\mathbb{E}(X^-)^{2+\delta} < \infty.$$

This stronger finite-variance-type condition is used only for direct LAWS expectile estimation. It is not part of the quantile-based plug-in route used in the contribution chapters.

2.2 Tail-index and extreme-quantile estimation

We collect the two classical estimators used throughout the thesis: the Hill estimator of γ and the Weissman extrapolation for extreme quantiles. The textbook reference is Haan and Ferreira (2006, Ch. 3–4).

2.2.1 The Hill estimator

Given an iid sample $X_1, \dots, X_n$ from F and an integer $k = k_n \in \{1, \dots, n-1\}$, the *Hill estimator* (Hill 1975) of γ based on the $k+1$ top order statistics is

$$\hat{\gamma}_n(k) = \frac{1}{k} \sum_{i=1}^k \log \left(\frac{X_{n-i+1:n}}{X_{n-k:n}} \right). \quad (2.1)$$

Since $U(t) \rightarrow \infty$, $X_{n-k:n} > 0$ with probability tending to one for intermediate k ; log-based estimators are understood on this event. For tail-index estimation one may also shift the data, but shifts must be undone for quantile or expectile estimation. The consistency result below is Haan and Ferreira (2006, Theorem 3.2.2); the normal limit is Haan and Ferreira (2006, Theorem 3.2.5) when $\rho < 0$, with the $\rho = 0$ endpoint covered by the $m = 1$ specialisation of Daouia, Padoan, et al. (2024, Theorem 1).

Theorem 2.6. *Assume $\mathcal{C}_2(\gamma, \rho, A)$ with $\gamma > 0$. Let $k = k_n \rightarrow \infty$ and $k/n \rightarrow 0$ as $n \rightarrow \infty$. Then $\hat{\gamma}_n(k) \xrightarrow{\mathbb{P}} \gamma$. If, in addition, $\sqrt{k} A(n/k) \rightarrow \lambda \in \mathbb{R}$, then*

$$\sqrt{k} (\hat{\gamma}_n(k) - \gamma) \xrightarrow{d} \mathcal{N}\left(\frac{\lambda}{1-\rho}, \gamma^2\right).$$

The choice of k controls the usual bias–variance trade-off; here it is treated as fixed by the data analyst.

2.2.2 Weissman extrapolation to extreme quantiles

The *Weissman extrapolation* (Weissman 1978) extrapolates from the intermediate-order statistic $X_{n-k:n}$ — which estimates the quantile $Q_X(1 - k/n)$ at the intermediate level $1 - k/n$ — to a quantile at a very-extreme level $1 - p$ with $p = p_n \rightarrow 0$ as $n \rightarrow \infty$. Plugging the Hill estimator into the first-order identity $U(ty)/U(t) \rightarrow y^\gamma$ gives the Weissman estimator

$$\hat{q}_n^*(1 - p \mid k) = \left(\frac{k}{np}\right)^{\hat{\gamma}_n(k)} X_{n-k:n}. \quad (2.2)$$

The theorem below, the Hill-estimator specialisation of Haan and Ferreira (2006, Theorem 4.3.8), covers the far-tail regime $np = o(k)$; the case $np = O(1)$ is very-extreme in the sense that only finitely many observations are expected beyond the target.

Theorem 2.7. *Assume $\mathcal{C}_2(\gamma, \rho, A)$ with $\gamma > 0$ and $\rho < 0$. Let $k = k_n \rightarrow \infty$, $k/n \rightarrow 0$, $p = p_n \rightarrow 0$ with $k/(np) \rightarrow \infty$ and $\sqrt{k}/\log(k/(np)) \rightarrow \infty$. Suppose $\sqrt{k} A(n/k) \rightarrow \lambda \in \mathbb{R}$. Then*

$$\frac{\sqrt{k}}{\log(k/(np))} \left(\frac{\hat{q}_n^*(1 - p \mid k)}{Q_X(1 - p)} - 1 \right) \xrightarrow{d} \mathcal{N}\left(\frac{\lambda}{1-\rho}, \gamma^2\right).$$

The log-denominator is the cost of extrapolation. Up to that factor the limit is the Hill limit, which is why Daouia, Padoan, et al. (2024, §2.3) pool Weissman estimators geometrically. This pooled-Weissman law is the source result against which the rebuilt expectile estimator will be compared.

2.3 Expectiles

Expectiles were introduced by Newey and Powell (1987) as asymmetric-least-squares analogues of quantiles. Their role as coherent and elicitable risk measures is developed in Bellini et al. (2014), Ziegel (2016), Gneiting (2011), and Ehm et al. (2016).

2.3.1 Definition

Let X be a real-valued random variable with $\mathbb{E}|X| < \infty$, and fix a level $\tau \in (0, 1)$. The τ -*expectile* of X is defined as

$$\xi_\tau(X) := \arg \min_{\theta \in \mathbb{R}} \mathbb{E}[\eta_\tau(X - \theta) - \eta_\tau(X)], \quad \eta_\tau(u) := |\tau - \mathbb{1}\{u \leq 0\}| u^2. \quad (2.3)$$

The subtraction of $\eta_\tau(X)$ makes the criterion finite under $\mathbb{E}|X| < \infty$ and does not affect the minimiser; at $\tau = 1/2$ one recovers the mean.

The convex objective (2.3) has a unique minimiser, and its first-order condition reduces to the implicit equation

$$\tau \mathbb{E}[(X - \xi_\tau(X))_+] = (1 - \tau) \mathbb{E}[(\xi_\tau(X) - X)_+], \quad (2.4)$$

where $u_+ = \max(u, 0)$. Thus expectiles balance upper and lower first partial moments, unlike quantiles, which balance tail probabilities.

2.3.2 Basic properties

The following properties are immediate from (2.3) or (2.4); we collect them without proof and refer to Newey and Powell (1987, §2) and Bellini et al. (2014, Sections 2–3) for details.

Proposition 2.8. *Let X and Y be real-valued random variables with $\mathbb{E}|X|, \mathbb{E}|Y| < \infty$. For each $\tau \in (0, 1)$:*

- (i) **Existence and uniqueness.** $\xi_\tau(X)$ exists and is uniquely defined.
- (ii) **Law invariance.** If X and Y have the same distribution, then $\xi_\tau(X) = \xi_\tau(Y)$.
- (iii) **Mean recovery.** $\xi_{1/2}(X) = \mathbb{E}X$.
- (iv) **Location equivariance.** $\xi_\tau(X + c) = \xi_\tau(X) + c$ for $c \in \mathbb{R}$.
- (v) **Positive homogeneity.** $\xi_\tau(\lambda X) = \lambda \xi_\tau(X)$ for $\lambda \geq 0$.
- (vi) **Sign reflection.** $\xi_\tau(-X) = -\xi_{1-\tau}(X)$.
- (vii) **Monotonicity in τ .** The map $\tau \mapsto \xi_\tau(X)$ is nondecreasing on $(0, 1)$, and is strictly increasing when X is nondegenerate, with the boundary limits $\xi_\tau(X) \rightarrow \text{ess inf}(X)$ as $\tau \rightarrow 0$ and $\xi_\tau(X) \rightarrow \text{ess sup}(X)$ as $\tau \rightarrow 1$ (infinite limits allowed).
- (viii) **Monotonicity in the argument.** If $X \leq Y$ almost surely, then $\xi_\tau(X) \leq \xi_\tau(Y)$.

For losses, expectiles are translation equivariant, positively homogeneous and monotone. The remaining coherence issue is subadditivity.

2.3.3 Coherence and elicibility

In the rest of the thesis X denotes a loss. The coherence literature is often written for financial positions, so in this subsection only write Y for a position and define the capital requirement

$$\rho_\tau(Y) = \xi_\tau(-Y).$$

Theorem 2.9 (Bellini et al. (2014)). *The functional $\rho_\tau(Y) = \xi_\tau(-Y)$ is a coherent risk measure if and only if $\tau \geq 1/2$.*

Recall that a functional is *elicitable* if it is the unique minimiser of an expected strictly consistent score (Gneiting 2011). Expectiles are elicitable on classes with finite first moment; the asymmetric squared score is strictly consistent under finite second moments, and the general scoring class has Bregman-type and Choquet-type representations (Gneiting 2011; Ehm et al. 2016). This scoring structure motivates forecast comparison, but the asymptotic theory below uses only (2.4) and the extreme expectile–quantile equivalence.

2.4 Extreme expectile inference: the iid heavy-tailed baseline

We now consider high expectile levels for a heavy-tailed loss with $\gamma \in (0, 1)$. Two regimes are used throughout:

- *intermediate-level*: $\tau_n \rightarrow 1$ with $n(1 - \tau_n) \rightarrow \infty$. There are still many observations above $Q_X(\tau_n)$, and central limit theorems hold at the rate $\sqrt{n(1 - \tau_n)}$.
- *extreme-level*: $\tau'_n \rightarrow 1$ with $n(1 - \tau'_n) \rightarrow c \in [0, \infty)$. There are at most finitely many observations above $Q_X(\tau'_n)$, and direct estimation is infeasible; extrapolation is required.

The iid results are due to Daouia, Girard, et al. (2018), Daouia, Girard, et al. (2019), and Daouia, Girard, et al. (2020); the plug-in formulation used later is adapted from the time-series extension of Davison et al. (2023).

2.4.1 The expectile–quantile asymptotic equivalence

The single most important fact for extreme expectile inference is that, in the heavy-tailed regime, the expectile is asymptotically proportional to the quantile of the same level.

Proposition 2.10 (Asymptotic equivalence; Bellini et al. (2014), Daouia, Girard, et al. (2019), Corollary 1 with $p = 2$). *Suppose X satisfies Condition 2.2 with index $\gamma \in (0, 1)$ and $\mathbb{E} X^- < \infty$; equivalently, suppose Condition 2.4 holds. Then*

$$\frac{\xi_\tau(X)}{Q_X(\tau)} \xrightarrow{\tau \uparrow 1} \psi(\gamma) := (\gamma^{-1} - 1)^{-\gamma}. \quad (2.5)$$

The bridge requires only the finite-mean range $\gamma < 1$ and lower-tail integrability; the stronger $\gamma < 1/2$ restriction enters direct LAWS CLTs, not the quantile-based route. The function ψ is

continuous on $(0, 1)$, with $\psi(\gamma) \rightarrow 1$ as $\gamma \downarrow 0$ and $\psi(\gamma) \rightarrow \infty$ as $\gamma \uparrow 1$. It is *not* monotone: its logarithmic derivative

$$m(\gamma) := \frac{d}{d\gamma} \log \psi(\gamma) = \frac{1}{1-\gamma} - \log(\gamma^{-1} - 1) \quad (2.6)$$

changes sign at $\gamma^* \approx 0.2178$, where ψ reaches its minimum. This non-monotonicity is relevant later when $m(\gamma) = 0$ makes some weight formulas degenerate.

Under the second-order condition $\mathcal{C}_2(\gamma, \rho, A)$ with $\gamma \in (0, 1)$ and $\mathbb{E} X^- < \infty$, the first-order equivalence (2.5) sharpens to the second-order expansion (Daouia, Girard, et al. 2020, Proposition 1(i))

$$\begin{aligned} \frac{\xi_\tau(X)}{Q_X(\tau)} &= \psi(\gamma) \left[1 + c(\gamma, \rho) A((1-\tau)^{-1}) + \gamma(\gamma^{-1} - 1)^\gamma \frac{\mathbb{E} X}{Q_X(\tau)} \right. \\ &\quad \left. + o(A((1-\tau)^{-1})) + o(Q_X(\tau)^{-1}) \right], \\ c(\gamma, \rho) &:= \frac{(\gamma^{-1} - 1)^{-\rho}}{1 - \gamma - \rho} + \frac{(\gamma^{-1} - 1)^{-\rho} - 1}{\rho}. \end{aligned} \quad (2.7)$$

At $\rho = 0$ the second term in $c(\gamma, \rho)$ is read by continuity, so $c(\gamma, 0) = m(\gamma)$. The A -term is the second-order expectile-quantile bias used later; the displayed first-moment term is $O(Q_X(\tau)^{-1})$ and is negligible under the moment-rate condition imposed in the QB CLT below.

2.4.2 Two families of estimators at intermediate levels

Two natural estimators of ξ_{τ_n} at an intermediate level τ_n arise from the asymmetric-LS definition (2.3) on the one hand, and from the asymptotic equivalence (2.5) on the other.

LAWS direct estimator. The *asymmetric-least-squares* (LAWS) estimator replaces the population minimisation (2.3) with its empirical analogue:

$$\tilde{\xi}_{\tau_n} = \arg \min_{\theta \in \mathbb{R}} \sum_{i=1}^n \eta_{\tau_n}(X_i - \theta). \quad (2.8)$$

Theorem 2.11 (Intermediate LAWS CLT; Daouia, Girard, et al. (2018), Theorem 2). *Assume Condition 2.2 with $\gamma \in (0, 1/2)$, and assume Condition 2.5. Let $\tau_n \rightarrow 1$ with $n(1 - \tau_n) \rightarrow \infty$. Then*

$$\sqrt{n(1 - \tau_n)} \left(\frac{\tilde{\xi}_{\tau_n}}{\xi_{\tau_n}} - 1 \right) \xrightarrow{d} \mathcal{N} \left(0, \frac{2\gamma^3}{1 - 2\gamma} \right).$$

Quantile-based plug-in estimator. A second family substitutes estimators of γ and of the intermediate quantile $Q_X(\tau_n)$ into the asymptotic-equivalence formula (2.5):

$$\hat{\xi}_{\tau_n} = \psi(\hat{\gamma}_n) \hat{Q}(\tau_n), \quad (2.9)$$

where $\hat{Q}(\tau_n) = X_{n - \lfloor n(1 - \tau_n) \rfloor : n}$ is the empirical quantile (an intermediate-order statistic) and $\hat{\gamma}_n$ is a tail-index estimator admitting the required $\sqrt{n(1 - \tau_n)}$ limit theory. The natural choice is the Hill estimator of (2.1) with effective sample size $k = \lfloor n(1 - \tau_n) \rfloor$.

Theorem 2.12 (Intermediate QB CLT; Daouia, Girard, et al. (2018), Corollary 2; Daouia, Girard, et al. (2020), Proposition 1). *Assume $\mathcal{C}_2(\gamma, \rho, A)$ with $\gamma \in (0, 1)$ and assume $\mathbb{E} X^- < \infty$. Let $\tau_n \rightarrow 1$ with $n(1 - \tau_n) \rightarrow \infty$ and suppose*

$$\sqrt{n(1 - \tau_n)} A((1 - \tau_n)^{-1}) \rightarrow \lambda \in \mathbb{R}, \quad \frac{\sqrt{n(1 - \tau_n)}}{Q_X(\tau_n)} \rightarrow 0.$$

With $\hat{\gamma}_n$ the Hill estimator at effective sample size $k = \lfloor n(1 - \tau_n) \rfloor$,

$$\sqrt{n(1 - \tau_n)} \left(\frac{\hat{\xi}_{\tau_n}}{\xi_{\tau_n}} - 1 \right) \xrightarrow{d} \mathcal{N} \left(\left(\frac{m(\gamma)}{1 - \rho} - c(\gamma, \rho) \right) \lambda, \gamma^2 (1 + m(\gamma)^2) \right),$$

with $m(\gamma)$ as in (2.6) and $c(\gamma, \rho)$ the second-order expectile-quantile constant of (2.7). The asymptotic bias has two pieces: the Hill bias $\lambda/(1 - \rho)$ propagated through the logarithmic derivative $m(\gamma)$ of ψ , and the deterministic expectile-quantile gap $-c(\gamma, \rho)\lambda$. The empirical quantile $\hat{Q}(\tau_n)$ contributes no asymptotic mean term at the $\sqrt{n(1 - \tau_n)}$ scale when centered at $Q_X(\tau_n)$; the condition $\sqrt{n(1 - \tau_n)}/Q_X(\tau_n) \rightarrow 0$ inherited from the theorem ensures that the moment-term contribution to the expectile-quantile gap is negligible.

This is the zero-first-moment-bias specialisation of Daouia, Girard, et al. (2018, Corollary 2), combined with Daouia, Girard, et al. (2020, Proposition 1). The imposed $\sqrt{n(1 - \tau_n)}/Q_X(\tau_n) \rightarrow 0$ removes the additional first-moment bias term present in the more general statement and is the finite-mean QB regime used in the contribution chapters.

Comparison. On the direct-LAWS range $\gamma \in (0, 1/2)$ neither LAWS nor QB uniformly dominates: their asymptotic variances cross at $\gamma \approx 0.2623$, and the LAWS variance diverges as $\gamma \uparrow 1/2$. The thesis uses QB because it composes with the pooled Hill and Weissman quantile primitives of Daouia, Padoan, et al. (2024), not because it is uniformly more efficient.

2.4.3 Weissman extrapolation to very-extreme expectile levels

For levels τ'_n so close to one that $n(1 - \tau'_n) \rightarrow c \in [0, \infty)$, neither $\tilde{\xi}_{\tau'_n}$ nor $\hat{\xi}_{\tau'_n}$ provides a consistent direct estimator of $\xi_{\tau'_n}$. Extrapolation from an intermediate level τ_n to the extreme level τ'_n is required.

The asymptotic equivalence (2.5) and the first-order identity $U(ty)/U(t) \rightarrow y^\gamma$ together yield the heuristic

$$\frac{\xi_{\tau'_n}}{\xi_{\tau_n}} \sim \frac{Q_X(\tau'_n)}{Q_X(\tau_n)} \sim \left(\frac{1 - \tau'_n}{1 - \tau_n} \right)^{-\gamma},$$

which suggests the *Weissman-type* extrapolated estimator

$$\bar{\xi}_{\tau'_n}^* = \left(\frac{1 - \tau'_n}{1 - \tau_n} \right)^{-\hat{\gamma}_n} \bar{\xi}_{\tau_n}, \quad (2.10)$$

where $\bar{\xi}_{\tau_n}$ is any consistent intermediate-level estimator satisfying a $\sqrt{n(1 - \tau_n)}$ -CLT, such as $\tilde{\xi}_{\tau_n}$ or $\hat{\xi}_{\tau_n}$ under the conditions above, and $\hat{\gamma}_n$ is a $\sqrt{n(1 - \tau_n)}$ -consistent tail-index estimator. We write $\tilde{\xi}^*$ and $\hat{\xi}^*$ for the LAWS-anchored and QB-anchored extrapolations, respectively.

Theorem 2.13 (Extreme-level CLT; Daouia, Girard, et al. (2018), Corollaries 3–4). *Assume $\mathcal{C}_2(\gamma, \rho, A)$ with $\gamma \in (0, 1)$ and $\rho < 0$, and assume $\mathbb{E} X^- < \infty$. Let $\tau_n \rightarrow 1$ satisfy $n(1 - \tau_n) \rightarrow \infty$ and let $\tau'_n \rightarrow 1$ satisfy $n(1 - \tau'_n) \rightarrow c \in [0, \infty)$, $(1 - \tau'_n)/(1 - \tau_n) \rightarrow 0$ and $\sqrt{n(1 - \tau_n)} / \log((1 - \tau_n)/(1 - \tau'_n)) \rightarrow \infty$. Suppose moreover that*

$$\sqrt{n(1 - \tau_n)} A((1 - \tau_n)^{-1}) \rightarrow \lambda \in \mathbb{R}, \quad \frac{\sqrt{n(1 - \tau_n)}}{Q_X(\tau_n)} \rightarrow 0.$$

If $\hat{\gamma}_n$ is the Hill estimator and $\bar{\xi}_{\tau_n}$ is the QB intermediate estimator of Theorem 2.12, then

$$\frac{\sqrt{n(1 - \tau_n)}}{\log((1 - \tau_n)/(1 - \tau'_n))} \left(\frac{\bar{\xi}_{\tau'_n}^*}{\bar{\xi}_{\tau'_n}} - 1 \right) \xrightarrow{d} \mathcal{N}\left(\frac{\lambda}{1 - \rho}, \gamma^2\right).$$

The same conclusion holds with $\bar{\xi}_{\tau_n}$ equal to the LAWS estimator of Theorem 2.11 under the additional direct-LAWS assumptions $\gamma \in (0, 1/2)$ and Condition 2.5.

The log-rate is the classical Weissman rate. At this scale the anchor estimator is first-order invisible, so the limit variance is γ^2 for both LAWS and QB anchors.

2.4.4 The plug-in central limit theorem

For pooled inference we need a version that treats $(\hat{\gamma}_n, \bar{\xi}_{\tau_n})$ as generic inputs. The following thesis-level lemma is adapted from Davison et al. (2023, Supplement, Proposition C.6); it uses a generic rate a_n and makes the deterministic extrapolation error explicit.

Proposition 2.14 (Plug-in CLT for extrapolated expectiles; adapted from Davison et al. (2023), Prop. C.6). *Assume $\mathcal{C}_2(\gamma, \rho, A)$ with $\gamma \in (0, 1)$ and $\rho < 0$, and assume $\mathbb{E} X^- < \infty$. Let $a_n \rightarrow \infty$, let $\hat{\gamma}_n$ be an estimator of γ , and let $\hat{\xi}_{\tau_n}^I$ be an intermediate-level estimator of ξ_{τ_n} satisfying the joint asymptotic representation*

$$a_n \begin{pmatrix} \hat{\gamma}_n - \gamma \\ \hat{\xi}_{\tau_n}^I / \xi_{\tau_n} - 1 \end{pmatrix} \xrightarrow{d} \begin{pmatrix} \Gamma \\ \Delta \end{pmatrix}, \quad (2.11)$$

for some random vector $(\Gamma, \Delta)^\top$. Let $\tau_n, \tau'_n \rightarrow 1$ with $n(1 - \tau_n) \rightarrow \infty$, $n(1 - \tau'_n) \rightarrow c \in [0, \infty)$, $(1 - \tau'_n)/(1 - \tau_n) \rightarrow 0$, put

$$\ell_n := \log\left(\frac{1 - \tau_n}{1 - \tau'_n}\right),$$

and suppose

$$\frac{a_n}{\ell_n} \rightarrow \infty, \quad a_n A((1 - \tau_n)^{-1}) \rightarrow \lambda_1 \in \mathbb{R}, \quad \frac{a_n}{Q_X(\tau_n)} \rightarrow \lambda_2 \in \mathbb{R}.$$

Assume moreover that $\mathbb{P}(\hat{\xi}_{\tau_n}^I > 0) \rightarrow 1$ and that the deterministic extrapolation error satisfies

$$\frac{a_n}{\ell_n} \{\log \xi_{\tau'_n} - \log \xi_{\tau_n} - \gamma \ell_n\} \rightarrow 0.$$

Then the Weissman-extrapolated expectile

$$\hat{\xi}_{\tau'_n}^* = \left(\frac{1 - \tau'_n}{1 - \tau_n} \right)^{-\hat{\gamma}_n} \hat{\xi}_{\tau_n}^I$$

satisfies

$$\frac{a_n}{\log((1 - \tau_n)/(1 - \tau'_n))} \left(\frac{\hat{\xi}_{\tau'_n}^*}{\hat{\xi}_{\tau_n}^I} - 1 \right) \xrightarrow{d} \Gamma.$$

Proof sketch. The statement is the log-scale Slutsky argument behind Davison et al. (2023, Supplement, Proposition C.6), written here as a background tool. The detailed thesis proof will be restored only if this plug-in route is used in the rebuilt main theorem. $\square$

Remark 2.15 (Deterministic ratio control). The deterministic log-ratio condition in [Proposition 2.14](#) is the high-level extrapolation-bias control

$$\frac{a_n}{\ell_n} \{ \log \xi_{\tau'_n} - \log \xi_{\tau_n} - \gamma \ell_n \} \longrightarrow 0.$$

In the iid heavy-tailed setting this control is supplied by the very-extreme expectile extrapolation argument of Daouia, Girard, et al. (2020, Theorem 5), invoked in Davison et al. (2023, Proposition C.6). It uses the sequential Weissman expansion along $y_n = (1 - \tau_n)/(1 - \tau'_n) \rightarrow \infty$ together with (2.7).

[Proposition 2.14](#) remains useful background, but the rebuilt argument must verify directly whether its speed mechanism applies to the pooled estimator eventually used in the contribution chapters.

2.5 The optimal pooling framework

We now recall the pooling framework of Daouia, Padoan, et al. (2024). The rebuilt thesis uses this framework as the source theory for pooled tail-index and extreme-quantile inference before applying the expectile-quantile bridge. The framework pools tail-index and extreme-quantile estimators from m samples, allowing heterogeneous sizes, second-order properties and cross-sample tail dependence.

2.5.1 Setup

Throughout [Section 2.5](#), m is fixed while $n = \sum_{j=1}^m n_j \rightarrow \infty$. Let $\mathbf{X}_i = (X_{i,1}, \dots, X_{i,m})^\top$, $i \geq 1$, be iid copies of $\mathbf{X} = (X_1, \dots, X_m)^\top$. Margin j is observed for $i = 1, \dots, n_j$ and has continuous right-heavy-tailed distribution F_j , tail quantile function $U_j = (1/\overline{F}_j)^\leftarrow$ and extreme-value index $\gamma_j > 0$; its order statistics are denoted $X_{1:n_j,j} \leq \dots \leq X_{n_j:n_j,j}$.

Tail inference in sample j uses its top $k_j + 1$ order statistics, where the effective sample size $k_j = k_j(n)$ satisfies $k_j \rightarrow \infty$ and $k_j/n_j \rightarrow 0$ as $n \rightarrow \infty$. The total effective sample size is

$k = \sum_{j=1}^m k_j$. Assume the asymptotic-proportionality conditions

$$\frac{n_1}{n_j} \rightarrow b_j \in (0, \infty), \quad \frac{k_1}{k_j} \rightarrow c_j \in (0, \infty), \quad (2.12)$$

with $b_1 = c_1 = 1$. The marginal F_j satisfies $\mathcal{C}_2(\gamma_j, \rho_j, A_j)$ (Condition 2.3). The thesis works under the *common-tail-index* assumption

$$(H_\gamma) \quad \gamma_1 = \dots = \gamma_m = \gamma.$$

Under this assumption, all common- γ vector limits below are expressed on the aggregate scale $\sqrt{k}$, with the constants c_j translating the local $\sqrt{k_j}$ rates.

The cross-sample asymptotic dependence is captured by a pairwise *tail-copula* structure (Daouia, Padoan, et al. 2024, §2.1).

Condition $\mathcal{J}(\mathbf{R})$ (tail-copula condition). For each pair (j, ℓ) with $j \neq \ell$, there exists a function $R_{j,\ell} : [0, \infty]^2 \setminus \{(\infty, \infty)\} \rightarrow [0, \infty)$ such that, for every $(x_j, x_\ell) \in [0, \infty]^2 \setminus \{(\infty, \infty)\}$,

$$\lim_{s \rightarrow \infty} s \mathbb{P}(1 - F_j(X_j) \leq x_j/s, 1 - F_\ell(X_\ell) \leq x_\ell/s) = R_{j,\ell}(x_j, x_\ell).$$

The case $R_{j,\ell} \equiv 0$ is asymptotic tail independence, implied by the independent samples used later in the thesis.

2.5.2 The pooled Hill estimator

Each sample contributes a Hill estimator $\hat{\gamma}_j(k_j)$ as in (2.1); we collect these into the random vector

$$\hat{\gamma}_n = (\hat{\gamma}_1(k_1), \dots, \hat{\gamma}_m(k_m))^\top.$$

Given a weight vector $\omega \in \mathbb{R}^m$ with $\omega^\top \mathbf{1} = 1$, the *pooled Hill estimator* is the weighted arithmetic combination

$$\hat{\gamma}_n(\omega) := \omega^\top \hat{\gamma}_n = \sum_{j=1}^m \omega_j \hat{\gamma}_j(k_j). \quad (2.13)$$

The affine constraint $\omega^\top \mathbf{1} = 1$ ensures consistency for the common γ under (H_γ) .

The fundamental asymptotic result is the joint CLT for the marginal Hill estimators.

Theorem 2.16 (Daouia, Padoan, et al. (2024), Theorem 1). *Assume $\mathcal{C}_2(\gamma_j, \rho_j, A_j)$ for every j , the tail-copula condition $\mathcal{J}(\mathbf{R})$, the proportionality conditions (2.12), and $\sqrt{k_j} A_j(n_j/k_j) \rightarrow \lambda_j \in \mathbb{R}$ for every j . Then*

$$(\sqrt{k_1}(\hat{\gamma}_1(k_1) - \gamma_1), \dots, \sqrt{k_m}(\hat{\gamma}_m(k_m) - \gamma_m))^\top \xrightarrow{d} \mathcal{N}(\mathbf{B}, \mathbf{V}),$$

with $\mathbf{B} = (\lambda_j / (1 - \rho_j))_{j=1}^m$ and $\mathbf{V}$ the symmetric matrix whose entries are

$$V_{j,\ell} = \begin{cases} \gamma_j^2, & j = \ell, \\ \frac{\gamma_j \gamma_\ell}{\max(b_j, b_\ell) \sqrt{c_j c_\ell}} R_{j,\ell}(b_j c_\ell, b_\ell c_j), & j < \ell. \end{cases}$$

The matrix is symmetric, so $V_{\ell,j} = V_{j,\ell}$ for $\ell > j$. Suppose, in addition, the common- γ condition (H_γ) . Then the same marginal estimators, now viewed on the common $\sqrt{k}$ scale, satisfy

$$\sqrt{k} (\hat{\gamma}_n - \gamma \mathbf{1}) \xrightarrow{d} \mathcal{N}(\mathbf{B}_c, \mathbf{V}_c),$$

where

$$(\mathbf{B}_c)_j = \sqrt{c_j \sum_{i=1}^m c_i^{-1} \frac{\lambda_j}{1 - \rho_j}} \quad j = 1, \dots, m.$$

The covariance matrix $\mathbf{V}_c$ is given by

$$(\mathbf{V}_c)_{j,\ell} = \left(\sum_{i=1}^m c_i^{-1} \right) \begin{cases} \gamma^2 c_j, & j = \ell, \\ \frac{\gamma^2}{\max(b_j, b_\ell)} R_{j,\ell}(b_j c_\ell, b_\ell c_j), & j < \ell, \end{cases}$$

with $(\mathbf{V}_c)_{\ell,j} = (\mathbf{V}_c)_{j,\ell}$ for $\ell > j$. Consequently, for any $\boldsymbol{\omega}$ with $\boldsymbol{\omega}^\top \mathbf{1} = 1$,

$$\sqrt{k} (\hat{\gamma}_n(\boldsymbol{\omega}) - \gamma) \xrightarrow{d} \mathcal{N}(\boldsymbol{\omega}^\top \mathbf{B}_c, \boldsymbol{\omega}^\top \mathbf{V}_c \boldsymbol{\omega}).$$

For independent samples, all off-diagonal entries vanish and $\mathbf{V}_c$ becomes the diagonal matrix

$$\mathbf{V}_c = \gamma^2 \left(\sum_{i=1}^m c_i^{-1} \right) \text{diag}(c_1, \dots, c_m). \quad (2.14)$$

Equation (2.14) is the covariance form to be used when the rebuilt main theorem is specialised to independent samples.

2.5.3 Variance- and AMSE-optimal weights

The relevant weights solve quadratic programmes under the affine constraint $\boldsymbol{\omega}^\top \mathbf{1} = 1$; unless stated otherwise, they need not be nonnegative.

Proposition 2.17 (Optimal weights; Daouia, Padoan, et al. (2024), Theorem 1; estimated-weight implementation in §2.2 and Corollary 1). *Suppose $\mathbf{V}_c$ is positive definite. Under the common- γ condition (H_γ) :*

(i) *The variance-optimal weights, minimising $\boldsymbol{\omega}^\top \mathbf{V}_c \boldsymbol{\omega}$ subject to $\boldsymbol{\omega}^\top \mathbf{1} = 1$, are*

$$\boldsymbol{\omega}^{\text{Var}} = \frac{\mathbf{V}_c^{-1} \mathbf{1}}{\mathbf{1}^\top \mathbf{V}_c^{-1} \mathbf{1}},$$

with corresponding asymptotic variance $\boldsymbol{\omega}^{\text{Var},\top} \mathbf{V}_c \boldsymbol{\omega}^{\text{Var}} = (\mathbf{1}^\top \mathbf{V}_c^{-1} \mathbf{1})^{-1}$.

(ii) The AMSE-optimal weights, minimising $(\omega^\top \mathbf{B}_c)^2 + \omega^\top \mathbf{V}_c \omega$ subject to $\omega^\top \mathbf{1} = 1$, are

$$\omega^{\text{AMSE}} = \frac{(1 + \mathbf{B}_c^\top \mathbf{V}_c^{-1} \mathbf{B}_c) \mathbf{V}_c^{-1} \mathbf{1} - (\mathbf{1}^\top \mathbf{V}_c^{-1} \mathbf{B}_c) \mathbf{V}_c^{-1} \mathbf{B}_c}{(1 + \mathbf{B}_c^\top \mathbf{V}_c^{-1} \mathbf{B}_c)(\mathbf{1}^\top \mathbf{V}_c^{-1} \mathbf{1}) - (\mathbf{1}^\top \mathbf{V}_c^{-1} \mathbf{B}_c)^2}.$$

In the independent-samples case, $\mathbf{V}_c$ is diagonal (Equation (2.14)) and the variance-optimal weights reduce to

$$\omega_j^{\text{Var}} = \left(\sum_{i=1}^m c_i^{-1} \right)^{-1} c_j^{-1} \sim \frac{k_j}{k}, \quad (2.15)$$

where the equivalence uses $c_i^{-1} = \lim k_i/k_1$. The finite-sample distributed version is

$$\tilde{\omega}_n^{\text{Var}} = (k_1/k, \dots, k_m/k)^\top,$$

which is exactly normalised and converges to ω^{Var} .

Feasible weights replace $\mathbf{V}_c$ and $\mathbf{B}_c$ by consistent plug-ins (Daouia, Padoan, et al. 2024, §2.2). If $\hat{\omega}^\top \mathbf{1} = 1$ exactly and $\hat{\omega} \xrightarrow{\mathbb{P}} \omega$, then $\hat{\gamma}_n(\hat{\omega})$ is $\sqrt{k}$ -equivalent to $\hat{\gamma}_n(\omega)$; the difficult practical step is stable estimation of $\mathbf{V}_c$ and the bias quantities.

2.5.4 Weighted geometric pooling of extreme quantiles

For estimation of an extreme quantile $q(1-p)$ with $p = p(n) \rightarrow 0$, each sample contributes a Weissman estimator

$$\hat{q}_j^*(1-p \mid k_j) = \left(\frac{k_j}{n_j p} \right)^{\hat{\gamma}_j(k_j)} X_{n_j - k_j : n_j, j}.$$

Because Weissman extrapolation is multiplicative, pooling is geometric on the original scale, equivalently linear on the log scale. It requires asymptotic equivalence of the marginal extreme quantiles:

Condition 2.18 (Tail homoskedasticity ($\mathcal{H}$)). For every $j \neq \ell$, $U_j(t)/U_\ell(t) \rightarrow 1$ as $t \rightarrow \infty$.

Equivalently, $Q_j(1-p)/Q_\ell(1-p) \rightarrow 1$ as $p \downarrow 0$. The pooled extreme-quantile estimator of Daouia, Padoan, et al. (2024, §2.3) is

$$\hat{q}_n^*(1-p \mid \omega) := \prod_{j=1}^m [\hat{q}_j^*(1-p \mid k_j)]^{\omega_j}. \quad (2.16)$$

Taking logs reduces pooling to weighted sums of the local identities $\log \hat{q}_j^*(1-p \mid k_j) = \log(k_j/(n_j p)) \hat{\gamma}_j(k_j) + \log X_{n_j - k_j : n_j, j}$; so the Hill part is handled by Theorem 2.16; (2.12) gives $\log(k_j/(n_j p)) = \log(k/(np)) + O(1)$.

Theorem 2.19 (Daouia, Padoan, et al. (2024), Theorem 2). Assume the hypotheses of Theorem 2.16, the common- γ condition (H_γ), the tail-homoskedasticity condition ($\mathcal{H}$), and, in addition, $\rho_j < 0$ for every $j = 1, \dots, m$. Let $p = p(n) \rightarrow 0$ with $k/(np) \rightarrow \infty$ and $\sqrt{k}/\log(k/(np)) \rightarrow \infty$.

Then for any ω with $\omega^\top \mathbf{1} = 1$ and any $j \in \{1, \dots, m\}$,

$$\frac{\sqrt{k}}{\log(k/(np))} \left(\frac{\hat{q}_n^*(1-p | \omega)}{Q_j(1-p)} - 1 \right) \xrightarrow{d} \mathcal{N}(\omega^\top \mathbf{B}_c, \omega^\top \mathbf{V}_c \omega).$$

The strict $\rho_j < 0$ assumption is part of the Weissman theorem. It also supplies the rate needed to recenter by any marginal $Q_j(1-p)$ under $(\mathcal{H})$ (Daouia, Padoan, et al. 2024, Supplement, Lemma B.2): for any j, ℓ ,

$$\log Q_j(1-p) - \log Q_\ell(1-p) = O(|A_j(1/p)|) + O(|A_\ell(1/p)|) = o(k^{-1/2}),$$

along the Weissman sequence.

Thus the pooled Weissman and pooled Hill limits share the same mean, variance and optimal weights up to the log-rate factor. This is the multi-sample analogue of the tail-index dominance in Theorem 2.13 and Proposition 2.14.

2.5.5 Tail-homogeneity testing and confidence intervals

The covariance structure $\mathbf{V}_c$ also yields a Wald-type chi-square test for the common- γ hypothesis (H_γ) , sometimes called *tail homogeneity* across samples.

Theorem 2.20 (Tail-homogeneity test; Daouia, Padoan, et al. (2024), Corollary 3). *Assume the hypotheses of Theorem 2.16, the null $H_0 : \gamma_1 = \dots = \gamma_m$, and the bias-free calibration $\lambda_j = 0$ for every j . The statistic*

$$\Lambda_n = k (\hat{\gamma}_n - \bar{\mu}_n \mathbf{1})^\top \widehat{\mathbf{V}}_c^{-1} (\hat{\gamma}_n - \bar{\mu}_n \mathbf{1}), \quad \bar{\mu}_n = \frac{\mathbf{1}^\top \widehat{\mathbf{V}}_c^{-1} \hat{\gamma}_n}{\mathbf{1}^\top \widehat{\mathbf{V}}_c^{-1} \mathbf{1}},$$

where $\widehat{\mathbf{V}}_c$ is a consistent estimator of the covariance matrix of $\sqrt{k}(\hat{\gamma}_n - \gamma \mathbf{1})$ under H_0 . Then $\Lambda_n \xrightarrow{d} \chi_{m-1}^2$ under H_0 and $\Lambda_n \xrightarrow{\mathbb{P}} \infty$ under fixed alternatives. A test of asymptotic size α rejects H_0 when $\Lambda_n > \chi_{m-1, 1-\alpha}^2$.

Daouia, Padoan, et al. (2024) state the bias-free version above. As a direct Gaussian consequence, rather than as an additional result stated in that paper, the same GLS calculation also shows that, if the asymptotic bias is retained rather than removed by undersmoothing or bias correction, the deviance has the noncentral limit

$$\Lambda_n \xrightarrow{d} \chi_{m-1}^2(\delta_\gamma), \quad \delta_\gamma = \mathbf{B}_c^\top \mathbf{V}_c^{-1} \mathbf{B}_c - \frac{(\mathbf{1}^\top \mathbf{V}_c^{-1} \mathbf{B}_c)^2}{\mathbf{1}^\top \mathbf{V}_c^{-1} \mathbf{1}}.$$

Thus the central chi-square limit holds whenever $\mathbf{B}_c \in \text{span}\{\mathbf{1}\}$, in particular under $\lambda_j = 0$ for every j . Under contiguous alternatives of order $k^{-1/2}$, the limit is likewise noncentral rather than divergent.

The same Gaussian limits give confidence intervals for the common γ and log-scale intervals for pooled Weissman quantiles; retained bias must be removed or included in the centering.

2.5.6 Distributed inference as a special case

In the independent common-distribution distributed setting, the samples are iid subdivisions of one common population. Then:

- the common- γ condition (H_γ) and the tail-homoskedasticity condition $(\mathcal{H})$ are trivially satisfied;
- the tail copula $R_{j,\ell}$ between any pair of distinct samples vanishes identically (independence implies tail independence);
- $\mathbf{V}_c$ in [Theorem 2.16](#) becomes the diagonal matrix [\(2.14\)](#);
- the variance-optimal weights of [Proposition 2.17](#) reduce to $\omega_j^{\text{Var}} \propto k_j$ as in [\(2.15\)](#).

Corollary 2.21 (Distributed oracle equivalence; Daouia, Padoan, et al. [\(2024\)](#), Corollary 6, Theorem 3 and Corollary 8). *Under the distributed-inference assumptions (iid across machines) of Corollary 5 in Daouia, Padoan, et al. [\(2024\)](#), suppose the sample fractions are asymptotically equal in the stronger sense $k_j/n_j = (k/n)(1+O(k^{-1/2}))$ for every j . Then, under the conditions of their Theorem 3, the variance-optimally-pooled Hill estimator $\hat{\gamma}_n(\tilde{\omega}_n^{\text{Var}})$ is $\sqrt{k}$ -asymptotically equivalent to the unfeasible benchmark Hill estimator $\hat{\gamma}_n^{(\text{Hill})}(k)$ built from the entire combined sample. If, in addition, $\rho < 0$ and $p = p(n)$ satisfies $k/(np) \rightarrow \infty$ and $\sqrt{k}/\log(k/(np)) \rightarrow \infty$, then the variance-optimally-pooled Weissman estimator $\hat{q}_n^*(1-p \mid \tilde{\omega}_n^{\text{Var}})$ is asymptotically equivalent to the unfeasible benchmark Weissman estimator built from the combined sample.*

[Corollary 2.21](#) is the efficiency benchmark the rebuilt pooled-expectile construction should preserve if the expectile-specific terms are negligible on the pooled Weissman scale.

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